

TUSHIT TIWARI

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SUMMARY

Full Stack Software Engineer and MERN Stack Intern at IIT Ropar (Vicharanashala Lab for Education Design) — 3rd-year Computer Science student at VIT Bhopal (CGPA: 8.1/10) with hands-on production experience designing, building, and shipping scalable web, mobile, and AI-powered applications. Proficient across the full development lifecycle — from REST API design and real-time systems to RAG pipelines, SQL & NoSQL databases, and cloud infrastructure (AWS EC2, S3, DynamoDB). AWS Certified Cloud Practitioner (CLF-C02) with a strong track record of delivering measurable impact in research and industry environments.

TECHNICAL SKILLS

- **Frontend:** React.js, Next.js 14, Flutter, HTML5, CSS3, Tailwind CSS, Framer Motion
- **Backend:** Node.js, Express.js, FastAPI, Django REST Framework, REST API Design, Authentication & Authorization, JWT, Google OAuth
- **Databases & AI:** Firebase, MongoDB, SQL (MySQL, SQLite), AWS DynamoDB, ChromaDB, TensorFlow, Scikit-learn, Pandas, OpenCV, RAG, Vector Embeddings
- **DevOps & Languages:** AWS (EC2, S3, DynamoDB, IAM), Docker, Git, CI/CD, Vercel, Render, Unity Engine, Java, C++, Python, JavaScript, TypeScript, Dart

PROJECTS

[AI Portfolio Assistant](#) | *React, FastAPI, ChromaDB, RAG, Groq LLM, Vercel*

- Designed a full-stack RAG-powered portfolio chatbot using ChromaDB vector database for semantic retrieval across 5 knowledge collections, reducing irrelevant responses by 40% versus keyword search.
- Built FastAPI backend with intelligent query routing using Groq API (Llama 3.1 8B); achieved under 800ms average response time with full CI/CD deployment across Vercel and Render.

[Spendy: Expense Splitting Mobile App](#) | *Flutter, Firebase, Dart, Cloud Functions*

- Developed a cross-platform bill-splitting app with real-time Firestore sync for groups of 10+ users, Google OAuth, FCM push notifications, and an analytics dashboard via FL Chart.
- Implemented UPI deep-linking to GPay, PhonePe, and Paytm pre-filling recipient and amount, reducing payment settlement from 5+ steps to 1 tap; sole developer managing full release lifecycle to production APK.

[Collab Sheets: Real-Time Collaborative Spreadsheets](#) | *Next.js 14, TypeScript, Firebase, Vercel*

- Engineered a multi-user spreadsheet for 10+ concurrent editors with live presence indicators, Firestore conflict resolution, and a custom formula engine with dependency graph and circular dependency detection.
- Reduced initial render time by 60% via cell virtualization; shipped CSV/JSON export, drag-to-resize, keyboard shortcuts; deployed via GitHub Actions CI/CD to Vercel in under 3 minutes per build.

[VR Table Tennis Game](#) | *Unity, C#, SteamVR, OpenXR, Physics Engine*

- Built an immersive VR game in Unity with real-time physics simulation, under 16ms collision detection, and a 3-level Finite State Machine AI opponent; published demo on YouTube with 200+ views.

WORK EXPERIENCE

MERN Stack Intern — AI & Full Stack, IIT Ropar VLED (Remote)

May 2025 – Present

- Developed full-stack features and REST APIs for ViBe (engagement platform) and PES (Peer Evaluation System), serving 500+ students and educators across 3 active courses at IIT Ropar.
- Integrated RAG-based AI modules into assessment tools, improving automated feedback relevance by 35% versus the prior rule-based system; collaborated in an Agile team of researchers and engineers.

Research Intern & SEO Strategist, TheGeostrata, New Delhi (Remote)

Apr 2024 – Oct 2024

- Planned and executed an SEO strategy that grew monthly organic page views by 30% in 6 months, tracked via Google Analytics and Search Console across 50+ optimized pages.
- Authored 10+ geopolitics research articles on Indo-Pacific strategy and defense policy, contributing to a 25% increase in returning visitors and measurable gains in domain authority.

LEADERSHIP & EXTRACURRICULARS

Core Member, E-Cell Design Team — VIT Bhopal University | 2023 – Present

Designed marketing creatives, pitch decks, and event branding for entrepreneurship events reaching 1,000+ students on campus.

EDUCATION

VIT Bhopal: B.Tech, Computer Science & Engineering (E-Commerce Technology)

Aug 2023 – May 2027

CGPA: 8.1 / 10.0

City Montessori School, Lucknow

Completed May 2023

High School: 93.8% | Intermediate: 86.25%

CERTIFICATIONS AND HONORS

AWS Certified Cloud Practitioner (CLF-C02) | Google IT Support Professional | MERN Full Stack, ETHNUS | Deep Learning, NPTEL IIT Ropar | Oracle Cloud AI Foundations & GenAI Professional | Wipro Earthian Award | Letter of Recommendation from Defense Minister of India